Conjecture 00000004858: the missing non-elementary hypothesis

lizaixi01, with OpenAI Codex assistance

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1 The necessary existence claim

The source defines a free subgroup as a rank-two nonabelian free subgroup and proposes a construction in a torsion-free hyperbolic group. It does not assume that the group is non-elementary. Under this literal universal construction claim, its necessary existence conclusion is false: the infinite cyclic group $(\mathbf{Z}, +)$ is torsion free and hyperbolic, but has no nonabelian free subgroup.

This refutation concerns that unqualified universal construction. It does not refute a statement restricted to non-elementary groups, or a conditional problem in which an existing rank-two free subgroup is already supplied. The proposed length and algorithmic bounds cannot repair the missing existence conclusion.

We use the standard word-hyperbolic definition: a finitely generated group is hyperbolic when its word metric satisfies Gromov hyperbolicity. For the Cayley-graph formulation and the distinction between elementary and non-elementary groups, see Druţu and Kapovich, *Geometric Group Theory*, author manuscript, Definition 11.120 and Example 11.121, printed p.407 (PDF p.427). In particular, the latter example includes free groups of rank one. We prove the required word metric calculation and hyperbolicity directly.

2 The genuine shortest-word metric

Take the finite generating set $\{1\}$ of the additive integer group, whose signed letters are 1 and -1. A word is an actual finite list $w = (a_1, \ldots, a_n)$ of these integers, evaluated by its integer sum. For every such word,

$$\left| \sum_{j=1}^{n} a_j \right| \leq n.$$

Conversely, every integer z is the sum of $|z|$ copies of its sign, with the empty word for $z = 0$. Thus the shortest word length is

$$L(z) = \min\{\text{length}(w) : a_j \in \{1, -1\},\ \text{sum}(w) = z\} = |z|.$$

Lean defines this length as the actual infimum of the displayed set, proves nonemptiness, the universal lower bound, and an attaining word. It then derives the equality, instead of assigning $|z|$ as the definition of word length. The actual Mathlib Cayley graph for $\{1\}$ has adjacency precisely $y - x = \pm 1$.

The associated word distance is consequently

$$d(x, y) = L(y - x) = |y - x|.$$

Lean also identifies its real coercion with the actual metric-space distance on $\mathbf{Z}$. In particular it is a genuine metric. For all four integers x, y, z, t, direct ordered arithmetic proves

$$d(x, y) + d(z, t) \leq \max\{d(x, z) + d(y, t),\ d(z, y) + d(x, t)\}.$$

This is Gromov's four-point condition with $\delta = 0$. For the precise metric-space definition, see Nica and Špakula, *Strong hyperbolicity*, arXiv:1408.0250v2, Definition 2.1, printed p.2. We cite only that metric-space definition: their later standing convention restricting hyperbolic *groups* to non-elementary ones is explicit on p.3, and is absent from the competition statement.

The integer group is torsion free: for every nonzero natural number n, the equation $nz = 0$ forces $z = 0$. The final formal counterexample collects finite generation, the actual shortest-word hyperbolicity, torsion freeness, and the subgroup obstruction for the same integer group. No geometric property is assumed merely by naming a predicate.

3 No actual free subgroup of rank two

The standard free group $F_2 = \text{FreeGroup}(\text{Fin}\, 2)$ has canonical generators a, b. Their reduced words show

$$ab \neq ba.$$

For any group homomorphism $f : F_2 \to (\mathbf{Z}, +)$, however, commutativity of the target gives $f(ab) = f(ba)$. Hence f is not injective.

The Lean target uses `Multiplicative` to express the actual additive integer group in multiplicative notation. It separately proves that, for every actual subgroup H of this group, there is no group isomorphism $F_2 \simeq H$: composing such an isomorphism with the subgroup inclusion would give the impossible injective homomorphism. Thus no pair of words, regardless of length or computation time, can produce the subgroup required by the literal universal claim.

4 Verification

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

The submission includes a new root build, transitive axiom audits for every theorem, and full official `leanchecker --fresh Main` replay with source-bound receipts. No `sorry`, `native_decide`, or additional axiom is used. OpenAI Codex assisted, and a separate supervising Codex agent reviewed source, report and PDF; this is AI review, not human certification. Organizer acceptance is pending.